EFFiton: Design and Implementation of Controller, User Interface, High-Voltage Current Source, PI Stirrer Controller of Electro-Fenton Filter Reactor

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***Abstract*— This paper presents the design and implementation of four core subsystems of an automated homogeneous electro-Fenton reactor for the degradation of methylene blue dye in textile wastewater: Power Supply, Microcontroller/FreeRTOS, User Interface, and Stirrer Control. A BJT MJ15024-based current source with LM358 op-amp feedback delivers controlled current of 0–200 mA on an 83 V high-voltage rail with perfect linearity (R^2 = 1.000) and zero leakage when inactive. FreeRTOS-based firmware on an ESP32-S3 implements a five-state machine that orchestrates all sensors and actuators in parallel through fourteen independent tasks. A user interface on a 7-inch touchscreen display provides parameter input, real-time sensor monitoring, and spectrophotometric data visualization over bidirectional JSON-based UART communication. A PI controller for the magnetic stirrer achieves zero steady-state error, a settling time of approximately 4 seconds, and a critically damped response over the 100–1200 RPM range. Of the 15 defined technical specifications, 13 were fully met, and the system successfully executed a complete electro-Fenton reaction demonstration on June 4, 2026. This platform demonstrates the feasibility of automating a laboratory-scale electro-Fenton reactor using affordable off-the-shelf components and open-source firmware.**

Keywords— Electro-Fenton, FreeRTOS, Current Source, PI Controller, User Interface, ESP32Introduction

The textile industry contributes IDR 218.2 trillion to Indonesia's GDP with an annual growth rate of 4.26% in 2024 [1][2]; however, it generates liquid waste containing synthetic dyes such as methylene blue, which are toxic, carcinogenic, and inhibit photosynthesis in aquatic organisms [3]. Ministry of Environment and Forestry Regulation No. 16 of 2019 sets the color limit for textile wastewater at 200 Pt-Co, while actual effluent can reach 1,073.47 Pt-Co [4][5]—requiring a degradation efficiency of at least 82%.

Previous treatment efforts using a combined adsorption-filter reactor with zeolite adsorption and photo-Fenton, developed by BRIN and UNS, achieved only 50% efficiency [6]. Furthermore, reliance on UV light sources introduces problems: the process is visually unobservable (black-box), requires large spatial footprint, and poses occupational safety risks, thereby limiting the potential for industrial-scale upscaling.

Group TA252601013 proposes replacing this approach with a homogeneous electro-Fenton (HEF) reaction combined with zeolite adsorption. In this process, H_2O_2 is produced in-situ through oxygen reduction at the cathode and the Fe^{2+} catalyst is regenerated electrochemically—entirely without UV [7]. The system architecture (B300) consists of a reagent vessel, seven peristaltic pumps, a main reactor, a spectrophotometer, a processing unit, a user interface, and a power supply, all designed so that the entire process flow runs automatically and on schedule.

This paper focuses on the design and implementation of four core automation subsystems: (1) Power Supply — BJT-based voltage-controlled current source and 12 V/5 V voltage distribution; (2) Microcontroller/FreeRTOS — FreeRTOS-based firmware on ESP32-S3 with a state machine, multi-sensor reading, and a communication protocol; (3) User Interface — a 7-inch touchscreen display with a complete interface for parameter input, real-time monitoring, and spectrophotometric visualization; and (4) Stirrer Control — magnetic stirrer speed control over 100–1,200 RPM with a PI controller based on Hall sensor feedback. These four subsystems are prerequisites for the overall automation of the electro-Fenton reaction and process control..

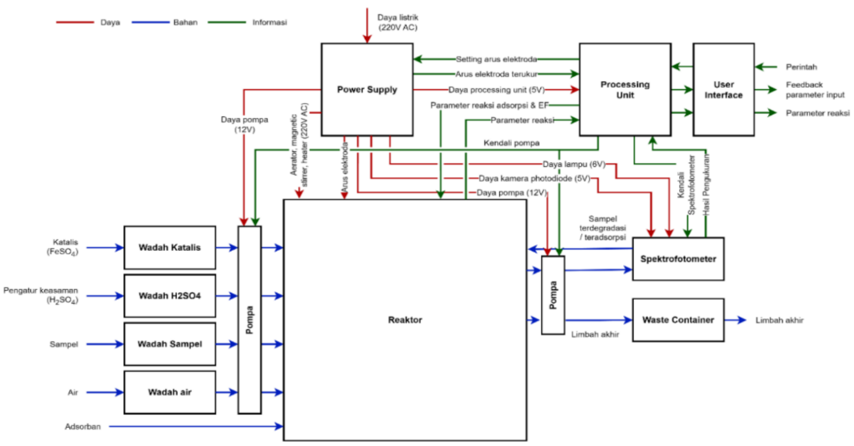

Figure 1 System Architecture-Adsorption & electro Fenton main system block diagram

I. LITERATURE REVIEW

1) *Proses Elektro Fenton*

The electro-Fenton process is an *Advanced Oxidation Process* (AOP) that exploits the Fenton reaction electrochemically to generate hydroxyl radicals (•OH) as the primary oxidizing agent. Unlike photo-Fenton processes that depend on UV irradiation, in electro-Fenton, H_2O_2 is produced *in-situ* via two-electron reduction of dissolved oxygen at the cathode, while Fe^{2+} as a catalyst is electrochemically regenerated from Fe^{3+} [7]. The principal reaction is:

$$\mathrm{Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^{-} + \cdot OH} \qquad (1)$$

The •OH radicals produced in equation (1) possess an oxidation potential of 2.80 V vs. SHE, making them one of the strongest oxidizing agents capable of effectively degrading persistent organic compounds including textile dyes [6]. The process is most effective under acidic conditions (pH 2–3) to maintain optimal Fe^{2+} catalyst activity. The elimination of UV dependency makes this process safer, more scalable, and better suited for industrial upscaling.

2) *Kendali PI Motor DC*

The speed of the DC stirrer motor is controlled using a closed-loop feedback loop based on a speed sensor. A Hall sensor generates a digital pulse each time a magnet mounted on the motor shaft is detected, enabling the rotational speed (RPM) to be calculated from the inter-pulse period. The selected controller is Proportional-Integral (PI), with the following control law:

$$u(t) = K_p \cdot e(t) + K_i \int_0^t e(\tau)\, d\tau \qquad (2)$$

In equation (2), $e(t)$ is the difference between the RPM setpoint and the measured RPM, K_p is the proportional gain, and K_i is the integral gain. The integral term theoretically guarantees *zero steady-state error*, while the derivative term is omitted because it would amplify *noise* in the Hall sensor pulse readings [9]. The parameters K_p and K_i are derived from the system transfer function model identified experimentally using the MATLAB System Identification Toolbox.

3) *Sistem Real-Time Berbasis FreeRTOS*

The reactor system requires sensor readings, actuator control, and data communication to occur simultaneously and in a timely manner. FreeRTOS as a *Real-Time Operating System* (RTOS) provides a *preemptive multitasking* mechanism that allows each function to be mapped to an independent task with configurable priorities and execution periods [8]. Shared resources between tasks (such as sensor variables and command queues) are protected using FreeRTOS built-in *mutexes* and *queues* to prevent *race conditions*. This approach ensures that the system's response to critical events (e.g., an operator-issued *force stop* command) is not delayed by other concurrently running operations.

II. ARCHITECTURE, DESIGN & SPECIFICATIONS

A. *Overall System Architecture*

This paper discusses four primary subsystems of the automated electro-Fenton reactor: (1) Power Supply, encompassing voltage distribution to all loads and a voltage-controlled current source for the reactor electrodes; (2) FreeRTOS-based Microcontroller serving as the main processing unit that orchestrates all sensors and actuators; (3) User Interface based on an HMI display enabling the operator to interactively control the reactor; and (4) Magnetic Stirrer Control with a closed-loop PI controller based on RPM feedback. These four subsystems are integrated to form the automation core of the reactor, enabling the execution of reaction schedules, real-time sensor monitoring, and an intuitive operator interface. All firmware source code is available at github.com/vincentDuckPilot/Fenton-upscale.

B. *Sub Sistem Power Supply*

1) *System Overall Power Supply Implementation*

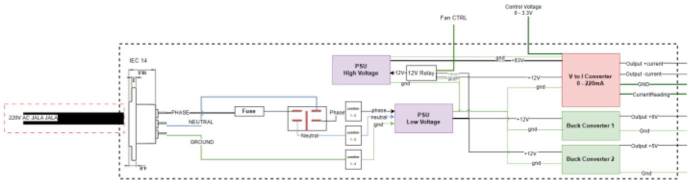

Figure 2 Power Supply Schematic

Table 1 Power Supply Specification

No	Specification	Value
1	Power Input	396.3 Watt

The power distribution system converts 220 V AC mains voltage into the various DC voltage rails required by the reactor subsystems, as illustrated in Figure 2. A 12 V/10 A main PSU serves as the power distribution backbone, from which a *boost* converter (12 V → 83 V) supplies high voltage for the electrode current source—fitted with a relay controlled by the controller to disconnect this rail when not needed. Three *buck* converters step down from the 12 V rail: two units (12 V → 6 V) supply peristaltic pumps and the spectrophotometer halogen lamp respectively, and one unit (12 V → 5 V) powers the ESP32, sensors, and display. All wiring uses Wago connectors (2-way, 3-way, and 6-way splitters) to minimize contact resistance and maintain installation neatness.

2) *Current Source Implementation*

Table 2 Current Source Specification

No	Specification	Value
1	Current	0 – 0.42A

A *voltage-controlled current source* (VCCS) is implemented using an LM358 op-amp with two paralleled MJ15024 power BJTs as the power amplification element. BJTs are chosen because their linear operating characteristics are better suited than switching MOSFETs under 83 V supply conditions with constant current up to 220 mA; the power dissipated by the BJTs, P = 83 V × 220 mA = 18.26 W, is well below the safe operating limit of the MJ15024. Two 2N2222 BJTs are used as active feedback to ensure current is evenly shared and to prevent *thermal runaway*, while emitter resistors R6 = R7 = 8.2 Ω serve as additional current limiters.

The control voltage ($V_{control}$) is supplied from the ESP32-S3 internal DAC pin (GPIO 25, 8-bit resolution, range 0–3.3 V). The relationship between $V_{control}$ and the output current I_{out} is derived from the op-amp *virtual-short* condition, yielding:

$$I_{out} = \frac{R_2}{R_5(R_1+R_2)} \cdot V_{control} \qquad (3)$$

With R1 = 680 kΩ, R2 = 51 kΩ, and R5 = 1 Ω, equation (3) gives $I_{out} \approx 0.0698 \cdot V_{control}$, such that at $V_{control} = 3.15$ V the result is $I_{out} \approx 220$ mA as specified. Current feedback is read by an ADS1115 (16-bit ADC, gain 16) measuring the voltage across R5 (1 Ω), where the voltage in mV is numerically identical to the current in mA. The relay on the 83 V rail prevents leakage current (~5 mA when $V_{control}$ = 0 V) when the current source is inactive, such as during the adsorption phase.

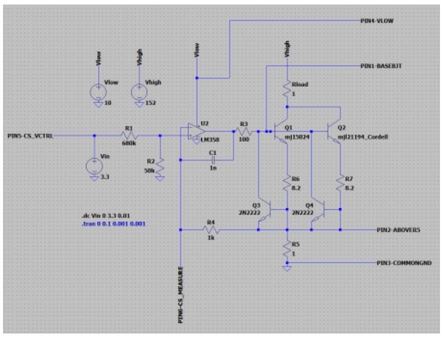

Figure 3 Current Source Circuit Design

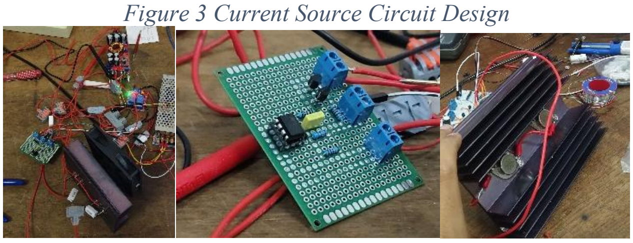

Figure 4 Current Source Circuit Implementation Photo. The most left is overall current source implementation, the midle is current source lower power path, the left is current source high power path

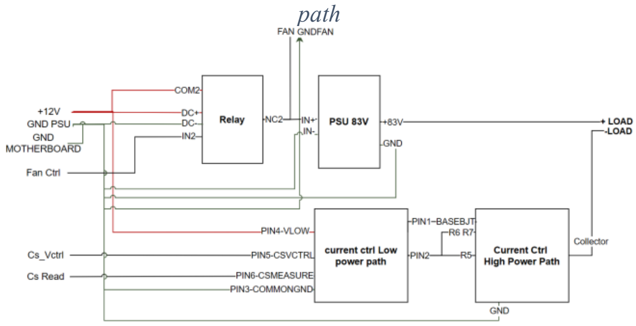

Figure 5 Current Source Circuit Layout

C. Microcontroller Sub System

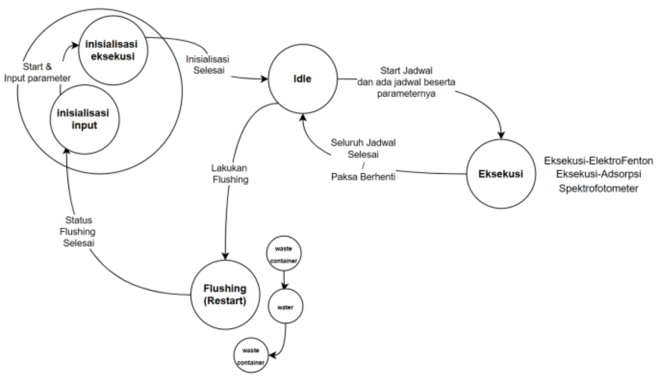

Figure 6 State Machine Firmware Controller

The processing unit uses an ESP32-DoitV1 as the main microcontroller. The ESP32-DoitV1 is selected because it supports *dual-core* operation, has an internal DAC for generating the current source control voltage (GPIO 25, 0–3.3 V), and is fully compatible with FreeRTOS on the ESP-IDF/Arduino framework..

FreeRTOS is used as a real-time operating system to manage multitasking on the ESP32. Each function is mapped to an independent task so that sensor, actuator, and communication operations run in parallel without blocking one another. Table II summarizes all implemented tasks and their functions.

Table 3 Task List

Subsystem Group	Task Name	Function
Reactor	taskBacaSensorSuhu	Read DS18B20 via OneWire protocol
Reactor	taskBacaSensorpH	Read ADS1115 ch. pH (PH1503C sensor)
Reactor	taskBacaSensorGas	Read ADS1115 ch. MQ-135 gas sensor
Reactor	interruptHall + taskBacaSensorHall	Calculate RPM via GPIO interrupt
Reactor	taskKontrolStirrer	PI controller for stirrer RPM
Reactor	taskKontrolHeater	On/off heater control
Reactor	taskTimer	Duration timing per reaction step
Power Supply	taskBacaSensorArus	Read ADS1115 ch. current source current
Power Supply	taskKontrolCurrentSource	Set DAC value → current source control voltage
Power Supply	taskKontrolCoolingFan	BJT heatsink fan control
Spectrophotometer	taskBacaSensorCahaya	Read TSL1401CL light sensor array
Spectrophotometer	taskKontrolHalogen	Halogen lamp on/off
Pump	taskManajemenPompa	Control 7 peristaltic pumps via PWM
Communication	taskKomunikasiUI	UART send/receive data to/from UI ESP32

The controller firmware implements a state machine with five operating states representing the complete experiment lifecycle, as illustrated in Figure 6. The Input Initialization state is the initial state in which the system waits for the operator to fill in parameters (sample volume, H_2SO_4, $FeSO_4$, and spectrophotometer calibration options) via the UI; all actuators are inactive. The Execution Initialization state activates pumps sequentially to fill the reactor and performs spectrophotometer calibration if selected, then automatically transitions to the Idle state. The Idle state is a standby condition where the controller continues to read sensors and send heartbeats to the UI, and permits the operator to compose a reaction schedule. The Execution state executes the schedule (EF, Adsorption, or Spectrophotometry) sequentially, with the current source only active during EF reactions, supports force stop commands, and returns to Idle upon completion. The Flushing state executes a three-stage reactor cleaning sequence—drain sample, fill with clean water, drain water—then resets all data and returns to Input Initialization.

D. User Interface Sub System

1) GUI Implementation

The user interface subsystem is implemented on a 7-inch Elecrow CrowPanel display (integrated ESP32-S3), communicating with the main controller ESP32-S3 via UART as two separate computing units. A PCF8674 I/O expander (I²C bus) connects a 4×4 matrix keyboard as an alternative input alongside the capacitive touchscreen, with * as the decimal separator and D as backspace. The interface design is created using EEZ Studio with the LVGL v8.4 framework, while the state transition logic is implemented in

C++ code on the display ESP32-S3. The UI state machine synchronizes with the firmware state machine and consists of four main states: Initialization (parameter form and fill progress monitor), Idle (reaction scheduling menu open, initialization menu locked), Execution (real-time sensor monitor, degradation chart updated every 5 minutes, Force Stop button available), and Flushing (cleaning progress monitor and chart data reset).

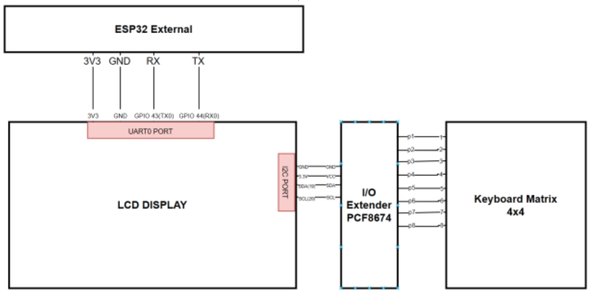

Figure 7 LCD Schematic Setup

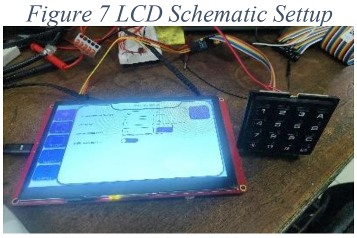

Figure 8 Schematic implementation

The interface implements six main menus: Welcome (splash screen and initial calibration option), Initialization (reagent volume input and pump progress), Scheduling (set schedule and parameters for each reaction), Monitor (all sensor readings in real-time), Spectrophotometry (absorbance spectrum chart and degradation chart per 5 minutes), and Flushing (cleaning progress and data reset).

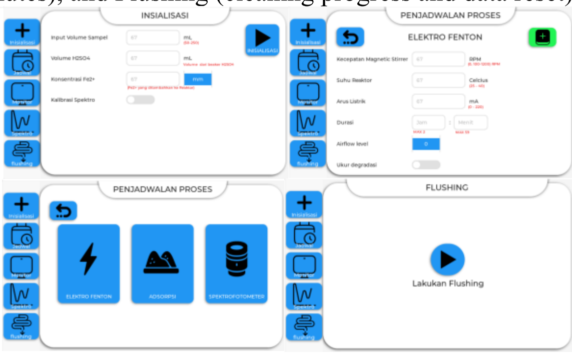

Figure 9 Snapshot UI layout designed in EEZ Studio

2) *Comunication Protocol*

The communication protocol between the UI and controller uses JSON format over UART. All parameter boundary validation (RPM, current, temperature, duration) is performed on the UI side before data is transmitted, so the controller can execute directly without re-validation. In the Initialization state, the UI sends a JSON packet containing vol, ph, fe_type, fe, and the kalibrasi flag; the controller responds with ACK_START, sends streaming status {"status":"ongoing",…}, and closes with ACK_DONE. In the Idle state, the controller sends a periodic heartbeat containing State, pH, suhu, and gas. In the Execution state, the UI sends the complete schedule in a single JSON packet ("status":"start", "jadwal":[…]); the controller responds with ACK_START and sends streaming sensor data containing fields reaksi, id, parameter, cs_on, and update_degradasi, with a dedicated 2-second delay for spectrum data transmission (401 points) to ensure data integrity; force stop is communicated via the message FORCE_STOP ↔ ACK:FORCE_STOP. In the Flushing state, the UI sends {"status":"flushing"} and the controller sends streaming progress until {"status":"flushing_done"}.

E. Pengendali Stirrer

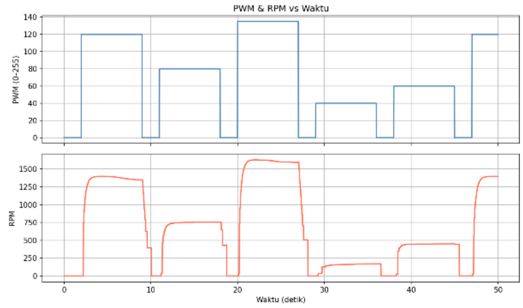

Figure 10 Input PWM vs RPM for system identification

The magnetic stirrer DC motor is controlled using an RPM feedback loop with a PI controller. The hardware consists of a DC motor with two permanent magnets, an LM393 Hall sensor that generates one LOW pulse per revolution, and an IRLZ44NPBF MOSFET-based H-Bridge (V_th = 1–2 V, compatible with ESP32 3.3 V signals) as the driver. RPM is calculated from the inter-pulse period within a 20 ms time window via GPIO interrupt, with the control signal as 100 Hz PWM at 8-bit resolution.
The mathematical model of the system is identified from experimental transient response data using the MATLAB System Identification Toolbox, yielding the second-order transfer function:

$$G(s) = \frac{24.41s + 0.41}{s^2 + 1.906s + 0.04642} \quad (4)$$

Based on the transfer function in (4), the PI controller is designed using the MATLAB PID Tuner with parameters K_p = 0.005 and K_i = 0.046, yielding a settling time of ~4 seconds with a critically damped response. The derivative term is not used because it would amplify *noise* in the RPM readings. The controller is implemented in the taskKontrolStirrer task on FreeRTOS, which reads the current RPM, computes the error against the *setpoint*, accumulates the integral, and updates the PWM value each iteration.

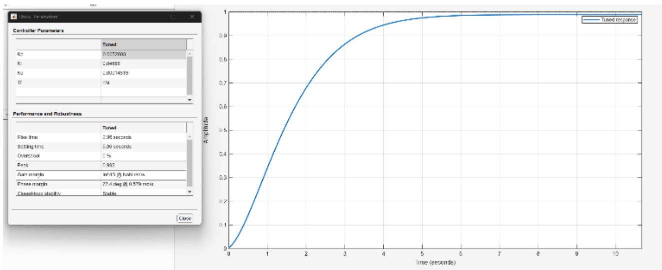

Figure 11 PI Matlab Tuner

F. \Sensor Gas dan Sensor pH

Several supporting sensors are calibrated to ensure adequate readings for the electro-Fenton process. The pH Sensor (PH1503C) is calibrated at the PPNN ITB laboratory using a Thermo Fisher (Toderlo) reference sensor in H_2SO_4 solutions of various concentrations (pH 2–5). The Gas Sensor (MQ-135) uses a *delta* voltage detection approach: a rise of

+0.1 V from the previous reading indicates the presence of CO_2, while a drop of −0.1 V indicates the absence of CO_2—this approach is used because the MQ-135 *baseline* voltage changes over time, making absolute readings unreliable.

III. Results and Analysis

A. Current Source

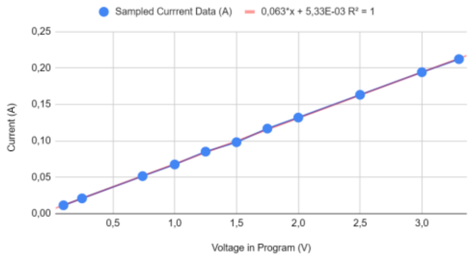

Figure 12 Program Voltage Input vs Current Actual

The *current source* subsystem was tested by varying the DAC output control voltage of the ESP32-S3 (GPIO 25) from 0 to 3.3 V, while measuring the actual output current through the load on the 83 V DC rail using an ADS1115 (gain 16) and verified with a digital multimeter. Figure 12 shows the linear relationship between the programmed DAC voltage (V_{input}) and the actual output current (I_{actual}):

$$I_{actual}(A) = 0.063 \times V_{input} + 0.00533 \qquad (5)$$

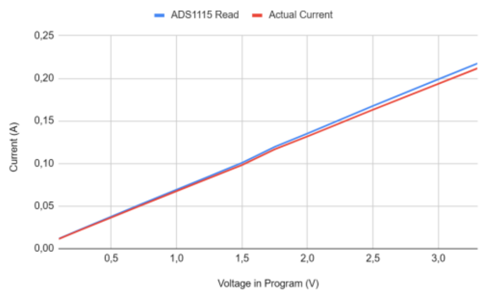

Figure 13 ADS1115 read vs Actual Current

A coefficient of determination R^2 = 1.000 confirms perfect linearity of the *current source* across the entire operating range (0–200 mA). AC ripple on the 83 V rail measured 0.003 V_rms — negligible for electrochemical applications. Additionally, zero leakage current was confirmed on the load side when the relay was commanded OFF during non-EF reaction phases. The *current source* was also fully integrated with the system schedule: it activates automatically during EF reactions and deactivates without leakage at the end, controlled via the relay and FreeRTOS task taskKontrolCurrentSource.

B. User Interface

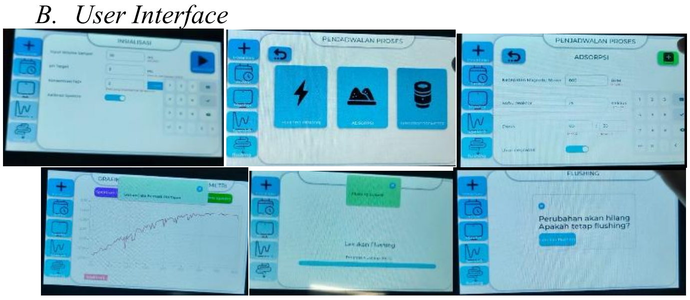

Figure 14 User Interface Implementation in Elecrow HMI dsiplay

The *user interface* subsystem is implemented on a 7-inch Elecrow CrowPanel display ESP32-S3 (800×480 px, capacitive touchscreen) running LVGL via EEZ Studio, and was verified through end-to-end functional testing covering all four states: Initialization, Idle, Execution, and Flushing. The following functional results were confirmed:

Table 4 UI Feature Checklist

Feature Tested	Result
UART communication UI ↔ Controller	✓ Operational, no data loss
Reaction schedule transmission (JSON-like format)	✓ Received and correctly executed by controller
Real-time sensor data display (RPM, temperature, current, pH, time)	✓ Updated in real-time
Force Stop command (UI → Controller → ACK)	✓ Confirmed and executed within one communication cycle
SD card data storage (CSV EF data, CSV spectro data)	✓ Data stored consistently
Multi-menu navigation (6 menus: Welcome, Initialization, Schedule, Monitor, Spectro, Flushing)	✓ All menus accessible at the correct state
Touchscreen responsiveness (after optimization)	✓ No lag after separating touch path from sensor polling
Keypad integration (4×4 matrix, * = decimal, D = backspace)	✓ Input correctly directed to focused field
Input validation (RPM 0 or 100–1200; Current 0–220 mA; Temperature 25–40°C)	✓ Out-of-range values rejected with warning pop-up

C. Stirrer controller PI configuration

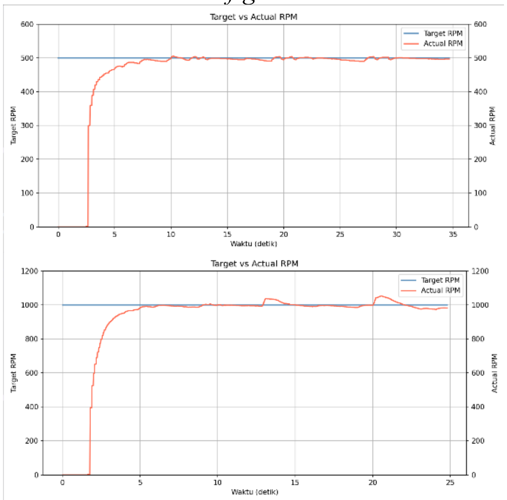

Figure 15 Step Response 500 RPM and 1000 RPM Set point

Based on the graph in Figure 15, the PI controller produces a steady-state error of 0 RPM thanks to integral action eliminating the permanent offset, with a measurement uncertainty of ±60 RPM due to the 20 ms window resolution of the Hall sensor. The settling time achieved is ~4 seconds with a critically damped characteristic — no overshoot was observed at either tested setpoint. The effective operating range is 100–1200 RPM; below 100 RPM, the 20 ms interrupt resolution is insufficient with only 1 pulse per revolution, making low-speed RPM readings inaccurate — this is a documented hardware limitation in the system specification. In full system integration testing (500 RPM and 1000 RPM for 5 minutes), the controller maintained the RPM target stably throughout the test duration.

D. pH sensor

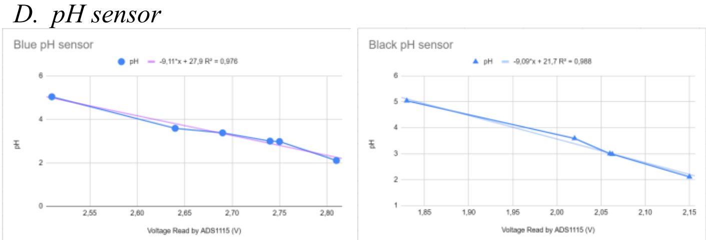

Figure 16 Voltage Read by ADS1115 vs pH

Calibrated at PPNN ITB using a Thermo Fisher (Toderlo) pH meter as reference, two PH1503C sensors (blue and black) were tested in four H_2SO_4 solutions (pH 2–5), yielding linear calibration equations (where x = sensor output voltage in volts):

$$\mathrm{pH}_{\mathrm{blue}} = -0.107 \times x + 3.05(R^2 = 97\%) \quad (6)$$

$$\mathrm{pH}_{\mathrm{black}} = -0.109 \times x + 2.39(R^2 = 97\%) \quad (7)$$

As shown in Figure 16, voltage noise reaches ±0.1 V (measured at ADS1115 GAIN_ONE), corresponding to a pH uncertainty of approximately ±0.011 pH for the given calibration slope. This level of accuracy is sufficient to monitor the target pH range of the electro-Fenton process (pH 2–3 for optimal Fe^{2+} activation)

IV. Conclusions

PThis paper has presented the design and implementation of four core subsystems of an automated electro-Fenton reactor: Power Supply, Microcontroller/FreeRTOS, User Interface, and Stirrer Control. The system was successfully integrated and fully demonstrated on June 4, 2026, with 13 out of 15 technical specifications fully met. Two specifications were partially met: the current source output current is limited to 200 mA (vs. 420 mA target) due to BJT thermal derating at the 83 V rail voltage, and stirrer control below 100 RPM is limited by the hardware resolution of the Hall sensor with a 20 ms window.

The MJ15024 BJT-based current source subsystem demonstrates perfect linearity ($R^2 = 1.000$) with a coefficient of 63 mA/V over the 0–200 mA range, AC ripple of 0.003 V_rms, and zero leakage current when the relay is OFF — all fully satisfying electrochemical process requirements. The stirrer PI controller successfully maintains RPM setpoints with zero steady-state error, a settling time of ~4 seconds, and a critically damped response without overshoot over the 100–1200 RPM range. The pH sensor was calibrated with R^2 = 97% accuracy over the pH 2–5 range, sufficient to monitor the optimal conditions for Fe^{2+} activation at pH 2–3.

At the system level, FreeRTOS-based firmware successfully implements a five-state machine that orchestrates all sensors and actuators in parallel, while the LVGL-based user interface can display real-time data and degradation charts without data loss over bidirectional UART communication. Future improvements include thermal derating optimization of the current source to achieve the 420 mA target, enhancement of low-speed RPM measurement resolution using a high-resolution encoder, and integration of automatic spectrophotometer calibration based on known wavelength standards to improve dye degradation monitoring accuracy.

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